Portfolio Optimization with Linear and Fixed Transaction Costs

Portfolio Optimization Project

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Based on the paper by Miguel Sousa Lobo, Maryam Fazel, and Stephen Boyd

Overview

1. Introduction
2. Problem Formulation
3. Transaction Costs
4. Constraints
5. Solution Methods
6. Project Extension
7. Conclusions

Introduction

Context

- Single-period portfolio optimization framework
- Key considerations:
  - Transaction costs (linear & fixed)
  - Risk constraints (variance, shortfall)
  - Portfolio constraints (diversification, shortselling)

Convex Problems

- Linear transaction costs
- Variance constraints
- Shortfall risk (Gaussian)
- ✓ Solvable via SOCP

Non-Convex Problems

- Fixed transaction costs
- Combinatorial nature
- × Requires heuristics
- Our contribution: Practical method

Portfolio Selection

Context

- The investor's goal is to maximize the expected wealth of n assets portfolio at the end of period, while satisfying a set of constraints on the portfolio.
- The current holdings in each asset are $w = (w_1, \ldots, w_n)$.
- The dollar amount transacted in each asset is specified by $x = (x_1, ..., x_n)$, with $x_i > 0$ for buying and $x_i < 0$ for selling.
- The portfolio budget contrainst is given by $\mathbf{1}^T x + \phi(x) = 0$, $\phi(x)$ the sum of all the transaction cost associated with x.
- The inequality form of the budget contrainst is more appropriate for use with numerical optimization methods (convexity). Notice the choice of the form equality or inequality lead to the same result.

Portfolio Selection Problem

Basic problems

$$
\begin{aligned}
\text{maximize} \quad & \bar{a}^T(w+x) \\
\text{subject to} \quad & \mathbf{1}^T x + \phi(x) \leq 0 \\
& w + x \in \mathcal{S}
\end{aligned} \tag{1}
$$

or

$$
\begin{aligned}
\text{minimize} \quad & \phi(x) \\
\text{subject to} \quad & \bar{a}^T(w+x) \geq r_{min} \\
& w + x \in \mathcal{S}
\end{aligned} \tag{2}
$$

Transaction Cost

Main assumption

The transaction costs is assumed to be separable, i.e., the sum of the transaction costs associated with each trade; $\phi(x) = \sum_{i=1}^{n} \phi_i(x_i)$

Linear Costs

$$\phi_i(x_i) = \begin{cases} \alpha_i^+ x_i & x_i \geq 0 \\ -\alpha_i^- x_i & x_i \leq 0 \end{cases} \tag{3}$$

- $\phi(x_i) = \alpha_i^+ x_i^+ + \alpha_i^- x_i^-$
- $\alpha_i^+, \alpha^- \geq 0$
- Convex set
- In pratice, transaction costs are not convex functions of the amount traded.

Transaction Cost

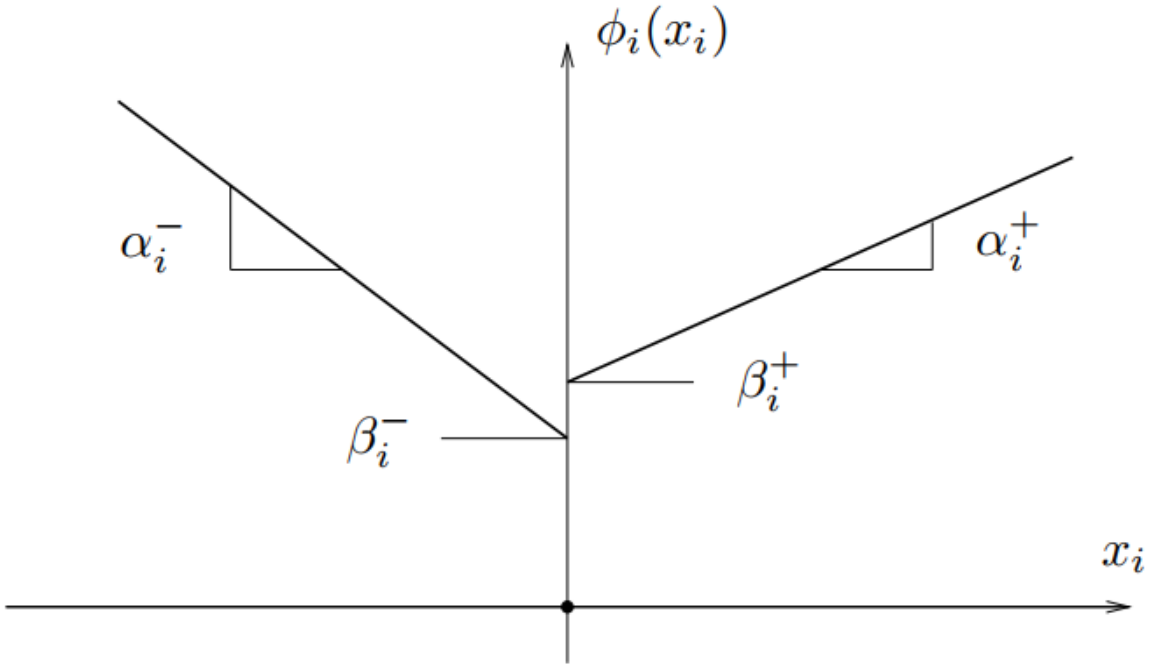
Fixed transaction cost

$$\phi_i(x_i) = \begin{cases} 0 & x_i = 0 \\ \beta_i^+ + \alpha_i^+ x_i & x_i > 0 \\ \beta_i^- - \alpha_i^- x_i & x_i < 0 \end{cases} \tag{4}$$

- Non-convex
- Requires special methods

Transaction Cost

Fixed transaction costs $\phi_i(x_i)$ as a function of transaction amount x_i



Portfolio constrainst

Diversification constrainst

- $$w_i + x_i \leq p_i$$
- This constraint limit the amount invested in each asset i to a maximum of p_i
- We can take $p_i = \gamma 1^T (w + x)$; γ a given proportion of the total ammount
- $$\sum_{i=1}^{r} (w + x)_{[i]} \leq \gamma \mathbf{1}^T (w + x)$$
- $f_{[i]}$ denotes the *ith* larger componant of f.
- This constrainst limits the maximum wealth held in any subset of r assets.

Portfolio constraints

Shortselling constraint

$$w_i + x_i \geq -s_i$$

- This inequality describe a limit on the maximum amount of shortselling allowed on asset i.
- In case of riskless asset, s_i is the credit line.

$$\sum_{i=0}^{n} (w + x)_i^- \leq S$$

- S is the total ammount of shortselling allowed.
- This constraint can be rewritten as :

$$\begin{aligned} -t_i &\leq w_i + x_i, \quad t_i \geq 0 \\ 1^T t &\leq S \end{aligned}$$

Portfolio constraints

Shortfall risk constraint

- Given $a \sim \mathcal{N}(\bar{a}, \Sigma)$, the shortfall risk constraint is defined by

$$P(W \geq W_{low}) \geq \eta$$

- $W = a^{\top}(w + x)$ is terminal wealth and W_{low} is the target floor.
- This constraint can be rewritten as :

$$\Phi^{-1}(\eta)||\Sigma^{1/2}(w + x)|| \leq \bar{a}^{T}(w + x) - W^{low}$$

- When $\eta \geq 0.5$, the problem is a convex optimization problem.
- The Markov and Chebyshev bound can be used to limit the shortfall probability.

Convex Problems

Convex optimization problem example

Consider the following problem

$$
\begin{aligned}
\text{maximize} \quad & \bar{a}^T(w + x^+ - x^-) \\
\text{subject to} \quad & 1^T(x^+ - x^-) + \sum_{i=1}^{n}(\alpha_i^+ x_i^+ + \alpha_i^- x_i^-) \leq 0 \\
& x_i^+ \geq 0, \quad x_i^- \geq 0 \\
& w_i + x_i^+ - x_i^- \geq -s_i \\
& \Phi^{-1}(\eta_j)\|\Sigma^{1/2}(w + x^+ - x^-)\| \leq \bar{a}^T(w + x^+ - x^-) - W_j^{\text{low}}
\end{aligned}
$$

- We condier100 risky and one riskless assets (so n $= 101$). This specifies linear transaction costs, a limit on shortselling of si per asset, and two shortfall risk constraints.

Convex Problems

Convex optimization problem example

$$w_1, \ldots, w_{100} = \frac{1}{101}, \quad w_{101} = \frac{1}{101}$$
$$\alpha_1^+, \ldots, \alpha_{100}^+ = 1\%, \quad \alpha_{101}^+ = 0$$
$$\alpha_1^-, \ldots, \alpha_{100}^- = 1\%, \quad \alpha_{101}^- = 0$$
$$s_1, \ldots, s_{100} = 0.005, \quad s_{101} = 0.5$$

$$\eta_1 = 80\%, \quad W_1^{\text{low}} = 0.9 \quad \text{et} \quad \eta_2 = 97\%, \quad W_2^{\text{low}} = 0.7.$$

- This problem is a convex optimization problem, with 202 variables and 306 constraints.

Non-convex problems (Fixed Costs)

Context

- We assume for simplicity on equal costs for buying and selling,the extension for nonsymmetric costs being straightforward.
- Transaction cost is then

$$\phi(x) = \sum_{i=1}^{n} \phi_i(x_i)$$

- We have

$$\phi_i(x_i) = \begin{cases} 0 & x_i = 0 \\ \beta_i + \alpha_i |x_i| & x_i \neq 0 \end{cases}$$

- Notice by considering the fixed costs, we discourage trading small amounts of a large number of assets. Thus, we obtain a sparse vector of trades; i.e., one that has many zero entries. This means most of the trading will be concentrated in a few assets, which is a desirable property.

Non-convex problems (Fixed Cost)

A first approach to solve

- **Method**: Exhaustive combinatorial search for the exact solution.
- **Per-asset choice**:
  - $x_i = 0 \Rightarrow \varphi_i(x_i) = 0$
  - $x_i \neq 0 \Rightarrow \varphi_i(x_i) = \alpha_i |x_i| + \beta_i$
- **Complexity**: 2^n possible combinations for n assets
- **Solution approach**:
  - Each combination yields a convex optimization problem
  - Global optimum maximize the objective of all 2^n solutions
- **Practical limits**:
  - Feasible for $n \leq 10$
  - Computationally intractable for $n > 20$
- **Solution**: Branch and Bound Method
  - Systematically eliminates suboptimal branches
  - Uses bounding to avoid full enumeration

Convex relaxation and global bound

Convex envelope

- We assume that lower and upper bounds on the x_i are known, i.e., l_i and u_i, for which x_i must satisfy $-l_i \leq x_i \leq u_i$.
- The convex envelope of ϕ_i, that we denote $\phi_i^{c \cdot e}(x_i)$, which is the largest convex function lower or equal to $\phi_i(x_i)$ in the interval $[-l_i, u_i]$, is given by

$$\phi_i^{c \cdot e}(x_i) = \begin{cases} \left(\frac{\beta_i}{u_i} + \alpha_i\right) x_i & ; \quad x_i \geq 0 \\ -\left(\frac{\beta_i}{l_i} + \alpha_i\right) x_i & ; \quad x_i \leq 0 \end{cases}$$

- Using $\phi_i^{c \cdot e}(x_i)$ instead of $\phi_i(x_i)$ relaxes the budget constraint, in the sense that it enlarges the search set.

Convex relaxation and global boud

Convex envelope

Consider the following optimization problem

$$\begin{aligned} \text{maximize} \quad & \bar{a}^T(w+x) \\ \text{subject to} \quad & \mathbf{1}^T x + \sum_{i=1}^{n} \phi_i^{c.e}(x_i) \leq 0 \\ & w + x \in \mathcal{S} \end{aligned} \tag{5}$$

- This corresponds to optimizing the same objective subject to the same portfolio constraints, as the initial problem, but with a looser budget constraint.
- The optimal value of (5) is an upper bound on the optimal value of the unmodified problem (with the $\phi(x)$ transaction cost fuunction)
- (5) is a convex problem, we can compute its optimal solution, and hence the upper bound on the optimal value of the original problem
- Notice that an ptimal solution for the original problem will be feasible for the modified problem (5), but not the opposite.

Convex relaxation and global bound

Convex envelope

The relaxation can also be used in the second form of the original problem,where the goal is to minimize the transaction cost.

$$\begin{aligned} \text{minimize} \quad & \sum_{i=1}^{n} \phi_i^{c.e}(x_i) \\ \text{subject to} \quad & \bar{a}^T(w+x) \geq r_{min} \\ & w + x \in \mathcal{S} \end{aligned}$$

- Compared to the original problem, the relaxed problem has the samefeasible set, but a different objective function.
- We can then obtain a lower bound for the optimal value of the original problem

Convex relaxation and global bound

Bound on the x_i

- Upper and lower bounds on the x_i are required to perform the convex relaxation and obtain a global upper bound on the expected end of period wealth
- The computation of such bounds depends on theparticular problem being addressed.
- For example, if the problem includes a constraint on the variance of portfolio return, an upper bound on x_i is given by the solution of

$$\begin{aligned} \text{maximize} \quad & f(x) = x_i \\ \text{subject to} \quad & (w+x)^T \Sigma (w+x) \leq \sigma^2 \\ & w + x \in \mathcal{S} \end{aligned}$$

- Forming the Lagrangian and minimizing, a simple calculation leads to:

$$x_i = \sigma \sqrt{(\Sigma^{-1})_{ii}} - w_i$$

- The same can be done for a lower bound.

Solving the portfolio optimization problem with fixed cost

Iterative heuristic

- **Key Idea**:
  - Suboptimal but feasible portfolio construction
  - Builds upon the relaxed problem solution method
- **Core Insight**:
  - Relaxed problem solution becomes feasible if:
    - $\phi^{c.e.}(x) = \phi(x)$
    - Transactions hit bounds (l_i, u_i) or zero
- **Modified Cost Function**:
  - Convex but not a lower bound
  - Reweighting drives small transactions to zero
  - Threshold δ determines zero tolerance
- **Properties**:
  - Each iteration solves a convex program
  - Maintains feasibility through agreement condition:
    - $\widehat{\phi}_i^k(x_i^*) = \phi_i(x_i^*)$

Solving the portfolio optimization problem with fixed cost

Iterative heursitic

1. $k := 0$. Solve the convex relaxed problem (5). Let x^0 be the solution to this problem.
2. $k := k + 1$. Given the solution to the previous problem x^{k-1}, define $\widehat{\phi}_i^k$ as

$$\widehat{\phi}_i^k(x_i) = \left(\frac{\beta_i}{|x_i^{k-1}| + \delta} + \alpha_i \right) |x_i|$$

Solve the modified (convex) portfolio selection problem

$$\begin{aligned} \text{maximize} \quad & \bar{a}^T (w + x) \\ \text{subject to} \quad & 1^T x + \sum_{i=1}^{n} \widehat{\phi}_i^k(x_i) \leq 0 \\ & w + x \in \mathcal{S} \end{aligned}$$

Let x_k be th solution of this problem.

3. If the portfolios x_k and x_{k-1} found in the two previous iterations are approximately equal, return $x^* = x_k$ and exit. Otherwise, go to step 2

Solving the portfolio optimization problem with fixed cost

Iterative heuristic

- If this iterative procedure exits, which occurs if two successive iterates are close to each other, the solution x^* will be nearly feasible for the original problem
- Because δ is relatively small, we have

$$\hat{\phi}_i(x_i^*) = \left(\frac{\beta_i}{|x_i^*| + \delta} + \alpha_i \right) |x_i^*| \approx \beta_i + \alpha_i |x_i^*| = \phi(x_i^*)$$

- Note that while $\hat{\phi}_i(x_i^*) \leq \phi_i(x_i^*)$ for all x_i^* , this inequality is tight except for x_i^* in the order of δ.
- On each iteration, for the x_i^k that become small, the modified $\hat{\phi}_i^{k+1}(x_i)$ has an increased derivative. This eventually pushes the small x_i to zero, leading to sparse solutions.

Solving the portfolio optimization problem with fixed cost

Iterative heuristic

- The same method is applicable to problem (2) , where we seek to minimize the transaction costs. The iterative heuristic is identical, except that, in each iteration instead of problem (5), the following problem is solved

$$\begin{aligned} \text{minimize} \quad & \sum_{i=1}^{n} \phi_i^k(x_i) \\ \text{subject to} \quad & \bar{a}^T(w+x) \geq r_{min} \\ & w + x \in \mathcal{S} \end{aligned}$$

- As a final step, an extra iteration can be included with the sparsity pattern fixed, and with the transaction costs exact for that pattern. That is, the small x_i are set to zero, with $\phi_i(x_i) = 0$, and the others are assumed nonzero, with $\phi_i(x_i) = \beta_i + \alpha_i|x_i|$
- The paper proves convergence for reweighting the objective in the cost-minimization formulation.

Fixed cost portfolio problems example and solving

Fixed cost problem example

$$
\begin{aligned}
w_i &= 1/11 \quad (i = 1, \ldots, 11), \\
\alpha_i &= 1\%, \quad \beta_i = 0.01 \quad (i = 1, \ldots, 10), \\
\alpha_{11} &= \beta_{11} = 0, \\
s_i &= 0.05 \quad (i = 1, \ldots, 10), \quad s_{11} = 0.5.
\end{aligned}
$$

- Example with 10 stocks, plus a riskless asset
- The small size of this problem is choosed to allows us compute the exact solution, that is the global optimum, by combinatorial search.

Fixed cost portfolio problems example and solving

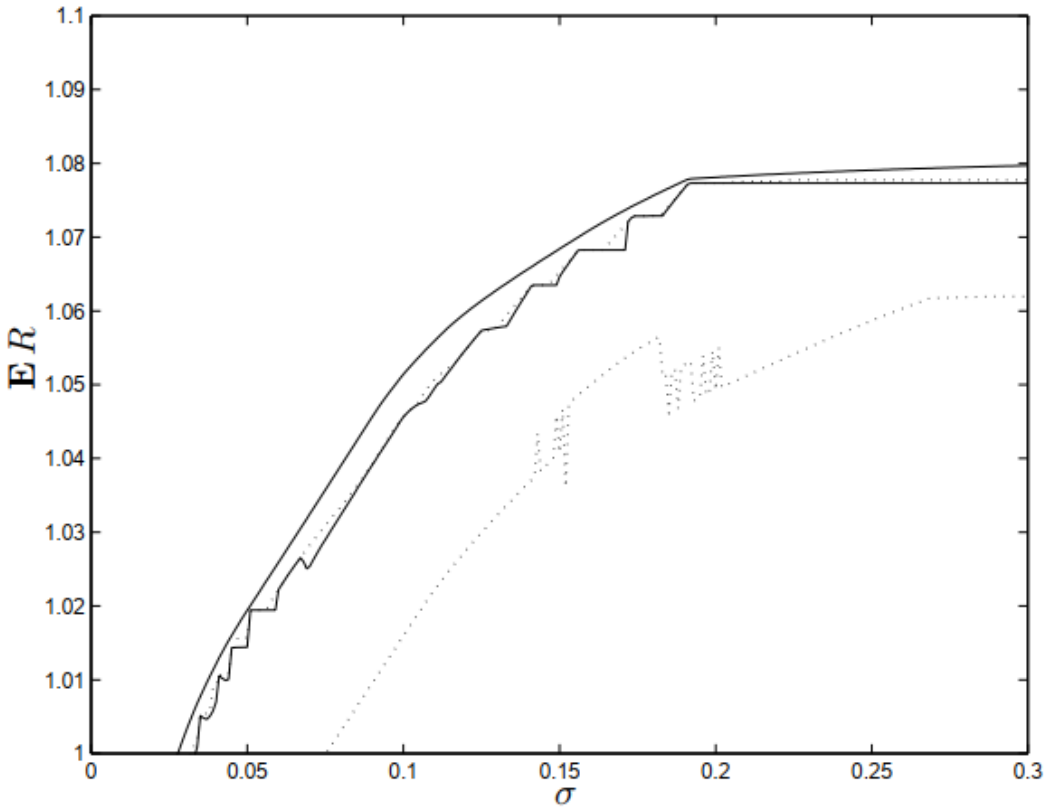


Figure: Expected return versus standard deviation for 10 stocks and a risk-free asset. From top to bottom, the curves show the upper bound, exhaustive optimum, heuristic, and solution that ignores fixed fees. The optimum and heuristic curves nearly coincide.

Screening Trades Before Exact Search

Directional upper bounds

Let L be the expected wealth of a portfolio that is feasible with exact fees. For asset i, compute an upper bound U_i^+ with an imposed buy and an upper bound U_i^- with an imposed sell.

- If $U_i^+ < L$, discard the buy direction. If $U_i^- < L$, discard the sell direction.
- If both inequalities hold, fix $x_i = 0$ before enumeration.
- A discarded direction can only yield wealth below an attainable value, so it cannot contain a global optimum.

Computational contribution

Reuse the relaxation's multipliers to bound all directions, keeping the exact fixed fees and part of the variance curvature.

Affine Screening and Directional Bounds

Let $b(x) = \mathbf{1}^\top x + \phi(x)$ be the exact budget residual. For each remaining convex constraint $g_j(x) \le 0$, take an affine minorant $\ell_j(x) \le g_j(x)$. For any feasible x and nonnegative multipliers λ, μ_j,

$$V(x) \le V(x) - \lambda b(x) - \sum_j \mu_j \ell_j(x) = c + \sum_i h_i(x_i), \qquad h_i(t) = s_i t - \lambda \phi_i(t).$$

For trades bounded by $l_i \le x_i \le u_i$, let $M_i = \sup_{l_i \le t \le u_i} h_i(t)$. Then

$$U_i^+ = c + \sum_{k \ne i} M_k + \sup_{0 < t \le u_i} h_i(t).$$

Use $l_i \le t < 0$ for the sell bound. The sum over the other assets is obtained by subtracting M_i from the precomputed total. This is the affine screening baseline.

Validity

The trade bounds must cover every feasible trade. The comparison value L must come from a feasible portfolio. The strict test $U_i^\pm < L$ preserves optima tied at L.

Tighter Bounds from Variance Curvature

Write the variance as $q(x) = (w + x)^\top \Sigma (w + x)$ with $q(x) \leq \sigma_{\max}^2$. Let $\hat{x}$ be the relaxed solution. Choose a diagonal $D \succeq 0$ such that $\Sigma - D \succeq 0$. Then

$$q(x) \geq q(\hat{x}) + \nabla q(\hat{x})^\top (x - \hat{x}) + (x - \hat{x})^\top D (x - \hat{x}).$$

This follows by expanding $q(x)$ and bounding its quadratic term below by the term with D.

- Subtract a nonnegative multiple of this lower bound for $q(x) - \sigma_{\max}^2$ in the directional bound. This preserves validity.
- The diagonal term adds a separate concave quadratic to each asset's contribution. Its maximum on either branch comes from projecting its vertex onto the trade interval.
- Keep the smaller of the affine and quadratic bounds for each direction. Both use the same relaxation.

Exact Search on 12 Risky Assets

Synthetic portfolio with one cash asset, symmetric fixed fees of 50 basis points of initial wealth per risky order, and proportional fees of 10 basis points of the order amount.

Method	Assets fixed	Patterns	Time
Full enumeration	0	4,096	44.61 s
Affine screening	0	4,096	—
Screening with curvature	2	1,024	13.02 s

Observed reduction

Fixing two assets reduces the search from 2^{12} to 2^{10} patterns. With seven preparation solves, the filtered run used 1,031 solves and was 3.43× as fast as full enumeration.

Single timed run per reported method, including filter preparation. Expected terminal wealth agreed within 3.36×10^{-10}.

Conclusions

- Convex problems efficiently solved with interior-point methods
- For fixed costs:
  - Heuristic provides good approximate solutions
  - Upper bound validates solution quality
- Directional screening can reduce exact search. The gain depends on the instance.
- Methods extend to related problems (index tracking)
- Future work: multi-period extensions

Thank you!

Questions?